

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B5: GENERAL RELATIVITY

TRINITY TERM 2025

Thursday 19 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

Expressed in terms of the Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$, the Riemann tensor has components

$$R^\gamma_{\alpha\beta\delta} = \frac{\partial}{\partial x^\delta} \Gamma_{\alpha\beta}^\gamma - \frac{\partial}{\partial x^\beta} \Gamma_{\alpha\delta}^\gamma + \Gamma_{\alpha\beta}^\epsilon \Gamma_{\delta\epsilon}^\gamma - \Gamma_{\alpha\delta}^\epsilon \Gamma_{\beta\epsilon}^\gamma.$$

The Ricci tensor is defined as $R_{\alpha\beta} \equiv -R^\gamma_{\alpha\beta\gamma}$. When expressed in a basis in which the metric tensor $g_{\alpha\beta}$ is diagonal, it can be written as

$$R_{\alpha\beta} = \frac{1}{2} \frac{\partial^2}{\partial x^\alpha \partial x^\beta} \ln |g| - \frac{\partial \Gamma_{\alpha\beta}^\gamma}{\partial x^\gamma} + \Gamma_{\alpha\gamma}^\delta \Gamma_{\beta\delta}^\gamma - \frac{1}{2} \Gamma_{\alpha\beta}^\gamma \frac{\partial}{\partial x^\gamma} \ln |g|,$$

where $|g|$ is the modulus of the determinant of $g_{\alpha\beta}$.

The Einstein equation is

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R - \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta},$$

where $R \equiv g^{\alpha\beta} R_{\alpha\beta}$ is the Ricci scalar, Λ is the cosmological constant and $T_{\alpha\beta}$ is the energy-momentum tensor.

For the line element

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right]$$

the Einstein equation can be written as the pair of equations

$$\begin{aligned} \ddot{R} &= -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) R + \frac{1}{3} \Lambda c^2 R, \\ \dot{R}^2 &= \frac{8\pi G}{3} \rho R^2 + \frac{1}{3} \Lambda c^2 R^2 - c^2 k, \end{aligned}$$

where ρ and p are the rest-frame density and pressure, respectively.

1. Consider a scalar function, $f(x^\mu)$, on a space-time with a metric $g_{\alpha,\beta}(x^\mu)$.
(a) Show that

$$\nabla_\mu \nabla_\nu f = \partial_\mu \partial_\nu f - \Gamma^\lambda_{\mu\nu} \partial_\lambda f .$$

[3]

- (b) Use the Jacobi formula

$$\partial_\lambda g = g g^{\mu\nu} \partial_\lambda g_{\mu\nu}$$

to show that

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{\sqrt{-g}} \partial_\lambda (\sqrt{-g}),$$

where g is the determinant of the metric.

[3]

- (c) Now show that

$$g^{\mu\nu} \nabla_\mu \nabla_\nu f = \frac{1}{\sqrt{-g}} \partial_\alpha (g^{\alpha\beta} \sqrt{-g} \partial_\beta f) .$$

[5]

- (d) Define

$$U_{\mu\nu} = \nabla_\mu \nabla_\nu f - g_{\mu\nu} \nabla^\lambda \nabla_\lambda f$$

to show that

$$\begin{aligned} \nabla^\mu U_{\mu\nu} &= R_\nu{}^\sigma \nabla_\sigma f \\ \nabla^\nu \nabla^\mu U_{\mu\nu} &= \frac{1}{2} \nabla^\nu R \nabla_\nu f + R_\nu{}^\mu \nabla^\nu \nabla_\mu f. \end{aligned}$$

You can use the fact that

$$(\nabla_\mu \nabla_\nu - \nabla_\nu \nabla_\mu) V^\alpha = R^\alpha{}_{\beta\mu\nu} V^\beta$$

where V^α is a 4-vector and the Bianchi identities are

$$\nabla_\gamma R^\mu{}_{\nu\alpha\beta} + \nabla_\beta R^\mu{}_{\nu\gamma\alpha} + \nabla_\alpha R^\mu{}_{\nu\beta\gamma} = 0 .$$

[8]

- (e) Show that, in a space-time with no matter and a cosmological constant, that

$$\nabla^\mu \nabla^\nu U_{\mu\nu} = \frac{1}{3} \Lambda g^{\mu\nu} U_{\mu\nu} .$$

[6]

2. The metric of a black hole in Schwarzschild coordinates is given by

$$ds^2 = - \left(1 - \frac{r_S}{r}\right) c^2 dt^2 + \left(1 - \frac{r_S}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2).$$

where $r_S = 2GM/c^2$ is the Schwarzschild radius.

(a) Derive the geodesic equations for a massive particle in this metric. Find an expression for the gravitation redshift of a signal emitted at $r = r_{\text{em}}$ and observed at $r = r_{\text{ob}}$, where r_{em} and r_{ob} are greater than r_S . What happens if $r_{\text{em}} = r_S$? [5]

(b) Find the proper time it takes for an object, released from rest at the horizon ($r = r_S$), to reach the singularity ($r = 0$).

Hint: $\int_0^1 \sqrt{\frac{x}{1-x}} dx = \frac{\pi}{2}$. [3]

(c) Consider two photons, travelling in the equatorial plane ($\theta = \pi/2$). They depart from a distance r_0 , towards a spherical object of radius $R \geq r_S$, centered at $r = 0$. Photon A is directed at $r = 0$ and photon B grazes the object at $r = R$ at which $\dot{r} = 0$. Show that, at r_0 , photon A has

$$(\dot{r}, \dot{\phi}) = ck(-1, 0),$$

and photon B has

$$(\dot{r}, \dot{\phi}) = ck \left(1 - \frac{r_S}{R}\right)^{-1/2} \left[\left(1 - \frac{R^2}{r_0^2}\right)^{1/2} \left(1 - \frac{r_S}{r_0}\right)^{1/2}, \frac{R}{r_0^2} \right].$$

[6]

(d) Use the unit-direction vectors of the velocities you found above to determine the angular size of the object for an observer at $r = r_0$, to show that the angular diameter, φ , of a star with radius R observed from a distance r_0 is

$$\varphi = 2 \arcsin \left[\frac{R}{r_0} \sqrt{\frac{1 - r_S/r_0}{1 - r_S/R}} \right].$$

[6]

(e) Sag A*, the black hole at the centre of our galaxy, has a distance $r_1 \simeq 8$ kpc from us and a mass of approximately $M_1 = 4.3 \times 10^6 M_\odot$, where $M_\odot$ is the mass of the Sun. Assume that the apparent size of the black hole is given by its light radius, $r_L = \frac{3}{2}r_S$ and find the angular size of Sag A*.

Hint: you may use the fact that the Schwarzschild radius of the Sun is $r_{S\odot} = 9.5 \times 10^{-17}$ kpc. [3]

(f) If, instead of using the light radius, we had used the Schwarzschild radius to define the physical size of the black hole, we would run into trouble defining the angular size. Explain why the expression diverges. [2]

3. The geometry of a wormhole is given by the line element,

$$ds^2 = -c^2 dt^2 + dr^2 + (R^2 + r^2)(d\theta^2 + \sin^2 \theta d\phi^2).$$

(a) Show that you can choose spatial slices such that invariant distance on those slices is given by

$$dS^2 = dr^2 + (R^2 + r^2)d\phi^2.$$

Define a new set of coordinates $z(r)$, $\rho(r)$ and find the conditions they must satisfy such that the metric can be written as

$$dS^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2.$$

[4]

(b) Show that a possible solution is

$$\rho(z) = R \cosh(z/R)$$

and sketch the 2-dimensional surface embedded in 3 dimensions. Explain why it is called a wormhole.

[3]

(c) Find the geodesic equations in this space time and determine the Christoffel symbols. Show that an object starting at coordinate $r = r_0$ and falling radially inward with a velocity V will take proper time

$$\Delta\tau = \frac{2r_0}{V}$$

to traverse the wormhole to coordinate $r = -r_0$.

[5]

(d) Consider the geodesic equations for a massless particle on the equatorial plane ($\theta = \pi/2$). Show that $u = 1/r$ obeys an equation of the form:

$$\frac{d^2 u}{d\phi^2} + u = 2R^2 \left[(\alpha R)^2 - 1 \right] u^3 + 2(\alpha R)^2 u$$

where α is a constant.

[7]

(e) Assume that $Ru \ll 1$ and assume the solution to the geodesic equation is of the form

$$u \simeq \frac{\sin \phi}{b_1} + \frac{\sin 3\phi}{b_2} + \frac{\phi \cos \phi}{b_3},$$

where b_1 , b_2 and b_3 are constants. Find expressions for these constants in terms of the impact parameter, b , of the geodesic in flat space ($R = 0$), $u_0(\phi)$, where

$$u_0 = \frac{\sin \phi}{b}.$$

Explain why there is no gravitational lensing in this space time.

Hint: $(\sin \phi)^3 = \frac{3}{4} \sin \phi - \frac{1}{4} \sin 3\phi$.

[6]

4. The metric for a homogeneous and isotropic universe is given by

$$ds^2 = -c^2 dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right),$$

where k is a constant.

(a) Using the geodesic equations, derive the Christoffel symbols for this metric. [5]

(b) Find Einstein field equations for a universe with a perfect fluid and a cosmological constant. [5]

(c) For this universe, derive an expression for the age of the universe, t_0 , as a function of the current values of the Hubble parameter H_0 , the density parameter for matter Ω_M , and the density parameter for the cosmological constant Ω_Λ . Find the age of a universe (in terms of H_0) in which:

(i) $\Omega_M = 1, \Omega_\Lambda = 0$;

(ii) $\Omega_M = 0, \Omega_\Lambda = 0$;

(iii) $\Omega_M = 0, \Omega_\Lambda = 1$.

Discuss what you found in (iii). [4]

(d) Discuss qualitatively the possible fates of a universe in the following scenarios:

(i) $\Omega_M > 1, \Omega_\Lambda = 0$,

(ii) $\Omega_M < 1, \Omega_\Lambda = 0$,

(iii) $\Omega_M = 1, \Omega_\Lambda > 0$. [3]

(e) Show that the angle subtended by an object of size R in the sky, placed at redshift z , is smaller in a slightly negatively curved universe ($k < 0$) as compared to in a slightly positively curved universe ($k > 0$). You may assume that the object has the same comoving distance,

$$D_c = \int_0^z \frac{c dz'}{H(z')},$$

in both universes.

Hint: recall that $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x)$ and $\int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arcsinh}(x)$. [6]

(f) The size of the typical hot spot and cold spot in the cosmic microwave background is set by the horizon at recombination or last scattering (at $z \simeq 1080$), which is roughly 147 Megaparsecs. Explain how you would use this fact and maps of the cosmic microwave background to infer the geometry of the Universe. [2]